

These could enable longer battery lives, or opt for higher screen resolution or opt for 5G cellular and so on. There are multiple things that our partners continue to innovate on that are beyond what we specify for the Evo brand.

Mithun: There have been quite a few criticisms regarding the wafer yield. How would address these concerns regarding supply from Intel to its partners?

Chris: Back during our earnings call, we've said that we have been shipping Tiger Lake chips for many weeks and we're ramping it strongly. We'll have 50 designs in the market by the end of this calendar year and it's a strong ramp and we're pleased with the ramp on the product.

Mithun: Just to be clear, you aren't going to be sourcing any part of Tiger Lake from an external fab, are you?

Chris: The Tiger Lake CPU and PCH are made in Intel factories and uses the Superfin technology.

Mithun: We've seen PCIe 4.0 on AMD IPs because their underlying Infinity Fabric is powered by PCIe and faster PCIe revision helps with performance, and they want to leverage PCIe 4.0 SSDs for faster data transfer. In Intel's case, how is PCIe 4.0 playing into power consumption, performance and cost to the end-user?

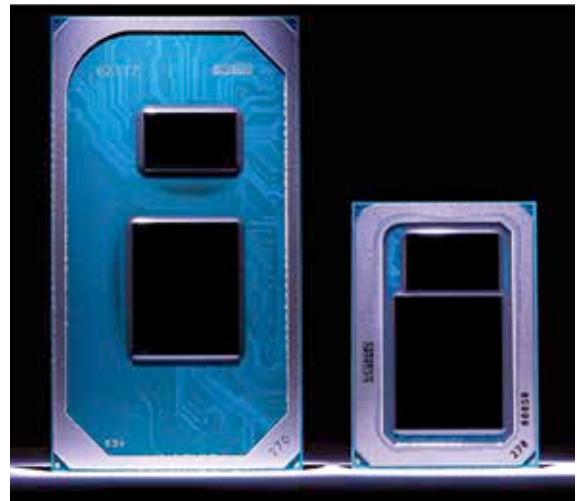
Chris: What we saw for PCIe Gen 4 is the ability to do things like high-speed

SSDs, direct CPU attach and discrete attach for the GPU, and it'll probably become more important in future products with the newer graphics cards that are coming next year. For us, it's a case of providing faster connectivity to key system elements.

Mithun: One of the outcomes of implementing PCIe 4.0 on motherboards was that the TDP of the PCH went up and the product cost to the end-user spiked. How would you address these concerns?

Chris: So the cost of PCIe Gen 4.0 devices, be it graphics or storage, is for those vendors and their strategies. For us, from a power standpoint, a lot of what Boyd talked about touched upon two things, the way power is managed across different domains and how we've implemented our ring architecture is how I believe we've effectively managed the total system power when we have a PCIe Gen 4.0 device connected.

Mithun: In your Blueprint series, you've gone over how AI has become ubiquitous and what I noticed was that most of the example tasks you've focused on were what creative professionals would indulge in. And in my view, creative professionals tend to



procure much more powerful systems such as a laptop with an Intel H-series processor and a discrete GPU. So I'm trying to understand what kind of AI-accelerated tasks would be more relevant for normal users?

Chris: The first thing that I would say is that for the kind of tasks that we spoke of, it's not just within the realm of professionals only. So when I think about creation in 2D or video, our belief is that it's not a niche thing. What happens with Adobe is that it used to be the domain of professionals but where AI comes in, it has automated a lot of the tasks and has reduced clicks and steps in their workflows. As I talk with end-users, more and more people are doing those creative workflows, not as high-end dedicated professionals but they're running their own businesses and the same laptop has to take them through the day whether it's meeting with a client or working on a video later on. So I think those workflows are and will be more broadly used by a lot of users. A lot of people are aspiring YouTubers and they graduate from what they can produce on their phone and then move up to what they can do on their laptops and desktops. I think it's more broad based and there are 12 million users of these tools. Also, AI is embedded in how we do power management and how we get that responsiveness. There's machine learning that helps us with the tuning and the governing of how long the system runs on Turbo boost frequency. So when we thought about AI, it was in the context of everyday uses to work in the background and to provide a cleaner experience. 

